

CNTb

A package for batch processing and statistical analysis of photon correlation spectroscopy data via CONTIN inversion

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I. INTRODUCTION

The CONTIN software, written by S. W. Provencher, allows one to analyze data that can be modelled by linear integral equations (see documentation in [CNTb/docs](#)). A canonical example is the auto-correlation function of the intensity scattered by particles with a size polydispersity, as measured by photon correlation spectroscopy. Written in FORTRAN in the beginning of the 80's, this very popular program has been implemented in several proprietary softwares of commercial equipments.

For practical reasons, it is often useful to analyze data independently of the measurement apparatus. Moreover, some problems can involve numerous experiments and it becomes tedious to analyze data one by one. Some batch processes allowing one to analyze hundreds of files in a single command line and to extract the results provided by CONTIN become then very useful. It is the purpose of the present scripts.

This directory contains the compiled CONTIN software together with the shell, Perl and Gnuplot scripts to use CONTIN on an Apple computer (OSX.5 and above) and to get the best of it. Only a basic knowledge of UNIX line commands is needed to use it. Other operating systems are supported as well (see section X below).

II. INSTALLATION

- In directory [CNTb/bin](#), keep the compiled CONTIN version that is consistent with your processor (PPC or Intel) et trash the other one. Change the name of the kept version to [contin](#).
- [CNTb](#) directory should be placed without any other modification in the Applications directory.
- [X11](#), [Gnuplot](#) and FORTRAN libraries should be installed (see IX).

III. USAGE

- Command lines are entered in a X11 terminal window.
- The usage is more convenient if the current working directory and the /Applications/CNTb directory are included in your environment. According to the default shell run in your session:
 - bash: edit or create the file ".bashrc" in [\\$HOME](#) directory and add:

```
PATH=./Applications/CNTb:$PATH    (mind the capitals !)  
export PATH
```

• csh: edit or create the file ".cshrc" in \$home directory and add:

```
set path = ( . /Applications/CNTb $path)    (mind the spaces !)
```

It is then no longer necessary to type the path to command `cnt`.

- There are two main usage modes: interactive (one file in a shot) or batch (one set or many sets of files in a shot). There is a third mode where a spectrum can be simulated and analyzed.

- Starting with version 4 (May 2013), the `CNTb` directory can be placed in any location you like. However, examples in this file assume that you located this directory in Applications folder.

REMINDER: NO SPACE ALLOWED IN DIRECTORY OR FILE NAMES !!!

IV. PRACTICAL USE

Create a working directory with the files to be processed by CONTIN, launch X11 and open a terminal window (Applications menu).

IV.1 Interactive mode:

• Go within the working directory with the UNIX command `cd`.

• Then type:

```
/A*/C*/cnt filename param
```

or, if it is ambiguous,

```
/Applications/CNTb/cnt filename param
```

and follow the instructions displayed on the screen (for `param` values, see below IV.6). In graphic mode, the script pauses after each display until the user presses any key or the mouse.

• When entering parameters for CONTIN, the user has to specify:

- the first and last data points to be considered by CONTIN: default = second and last points,

- the minimum and maximum values of relaxation times: no default values !,

- the need to preform or not the data (see RUSER10 in VII): default = do preform,

- the type of baseline, free or fixed (= 0): default = free baseline,

- the weight of residuals (weighted or unweighted): default = weighted residuals.

These values are saved in a file named `INFO_C` for optional use in the next run.

• After CONTIN completion, the user is prompted for options: o, f, r, z. The corresponding UNIX scripts "`o`", "`f`", "`r`" et "`z`" are executed by the line command `./x` (or merely `x` if the current directory has been added in the search path):

- "`o`" allows one to modify, through the UNIX command `vi`, the input parameters of CONTIN on the same data directly in the input file formatted for CONTIN. An other possibility is to modify this input file with a text editor and to quit `vi` without modification.

- "`f`" allows one to fit the distribution of relaxation times provided by CONTIN. The user

has then to enter the number of log-normal peaks in the fitting function, initial values for their parameters (amplitude, position and width) as well as the starting and ending values of the fitted region. These values are saved in a file named `INFO_F` and can be reused later, if the user enters a null string when asked for the number of peaks.

- "`r`" allows one to rename the temporary plots output during the analysis to prevent overwriting or deletion.
- "`z`" delete all the temporary files output during the analysis.
- When fitting the distribution function, the initial input parameters used in a previous fit can be modified with a text editor (see file `INFO_F` in VIII). Then press RETURN when the script asks for the number of peaks and the new values will be used for the next fit.

IV.2 Batch mode:

- This mode requires original `.ASC` data files.
- Starting Feb. 2013, version 2 of CNTb will explore recursively ALL subdirectories in the target directory to check if they contain data files. These data will then be automatically processed if one `INFO_C` file (mandatory) and one `INFO_F` file (if a fit is wanted) are included at the same level as the data files. Then CNTb does not ask for the input of CONTIN and distribution fit parameters but uses the values saved in these INFO files (see VIII for their format). If missing, CNTb asks once for this information and creates them.

NB: `INFO_C` and `INFO_F` can be created automatically both in interactive and batch modes.

- Command: `/A*/C*/cnt dirname param [f]`
where `dirname` is the full name (including path from the current working directory) of the directory to be analysed.
- When `param = p`, a pdf file containing plots of intensity traces and correlation functions is produced, which allows an easy one glance check of data quality. No CONTIN analysis. Bad quality files are marked on the plots and moved to the `_discarded` subdirectory. The default criteria defining bad quality can be modified by providing a file `INFO_P` (see VIII). Data can be restored to their place by setting `param = u`.
- Upon using "`f`" as third parameter, `/A*/C*/cnt dirname param f`
the script fits the distribution function for each file, saves the result in a file named `"*_fit.txt"`, and calculates statistics on the fitting parameters for the whole set of files in the directory. These statistics are saved in a file named `dirname.txt`. Also a weighted average correlation function is calculated from all the spectra, saved in the ALV format in the file `avrg_dirname.ASC` and analyzed in the same way as the individual spectra. All the experimental parameters should be identical for these operations to be meaningful. A routine checks this consistency and adds a warning at the end of this file in case of anomaly. A file `dirname.pdf` is created: it contains all the main plots generated during the analysis and allows their comparison in a glance. Inconsistent analyses can then be pointed. To remove them from the statistics, simply move the corresponding `.ASC` files from the `_ASC` subdirectory to the `_discarded` subdirectory (created automatically) and then enter the command:
`/A*/C*/cnt dirname s`
The statistics and the pdf files will be updated to contain only the information belonging to the files kept in `_ASC` subdirectory.

NB: for obvious reasons, the average correlation function is not recalculated. If the user suspects that

inconsistent results are not only a matter of fluctuations in CONTIN analysis but that the discarded spectra have no physical meaning, then the process should be started again from the beginning while keeping only the physically significant spectra. This task can be done mostly automatically by running first `cnt` with the param `p` after setting the values in `INFO_P` to meaningful values.

- The distributions of relaxation times can be fitted with values in `INFO_F` without running CONTIN with the command line: `/A*/C*/cnt dirname r`
- The mean values and standard deviations of the parameters found in the fitting procedure can be picked by typing the command line: `/A*/C*/cnt dirname e`
This will collect the results in all `subdirname.txt` files contained in `dirname` and save them in a new text file named `dirname_final.txt`.
- The info files `INFO_C` and `INFO_F` can be placed recursively in all appropriate subdirectories by typing: `/A*/C*/cnt dirname i`
provided a copy of them is placed in `dirname`. If not present, these copies will be created first by the scripts with the user input.
- By typing: `/A*/C/cnt dirname d`
all the .eps files generated by the scripts will be deleted, which can save a lot of space.
- By typing: `/A*/C/cnt dirname g`
all the .eps files generated by the scripts will be restored.

NB: if the environment was modified as explained above, there is no need to indicate the path to `cnt` BUT:

-) in interactive mode, the file to process has to be in the current working directory,*
-) in batch mode, the path from the working directory to the target directory should still be given !*

Starting March 2013, version 3 automatically discards bad fits of the distribution of relaxation times by Gnuplot. Badness criteria are the following:

The automatic peak search could not define the number of peaks requested by the user.

The fit gave unphysical negative values for one amplitude (typically ill-defined problem).

The error bar associated with one parameter exceeded a critical value (typically 50%).

The fit converged too slowly, using more than 30 iterations.

The fit stopped because the matrix inversion was impossible (determinant ≈ 0).

In all these cases, a suitable warning is printed in the corresponding "`*_fit.txt`", the fits are not saved, they are not considered in the statistical analysis and they are not plotted.

The plots of the distributions are labeled with a large colored "X" and with the warning.

IV.3 Simulation mode:

- One interesting feature of CONTIN was the possibility for the user to generate simulated data with a given level of noise and to test the ability of the program to recover the input distribution values. However the feeding of the program with these input values is nowadays not convenient for the average user. The easy use of this feature has been restored by appropriate scripts.

- By typing

`/A*/C*/cnt filename 3 B`

data are simulated with a noise level $B^{-1/2}$ (typ. $10^2 < B < 10^7$), stored in the form of a pseudo ALV file and then analyzed in interactive mode. A set of these files can later be reprocessed in batch mode as usual data. The input data are entered in the same form as the trial values for the fit of the distribution function.

IV.4 Summary of values for `param`:

- Interactive mode: $0 \leq \text{param} \leq 4$
 - Batch mode: `param` = 1, 2, s, p, r, e, i, d, g, u
- `param` = 0: no formatting of the data (the file is already an input file for CONTIN)
- `param` = 1: formatting
- `param` = 2: add 1 to data + formatting
- `param` = s: data file selection mode (CONTIN is not run and new statistics are calculated)
- `param` = p: raw data plot mode (only plot intensity traces and correlation functions)
- `param` = r: refit distribution functions produced by CONTIN.
- `param` = e: extraction of results saved in text files
- `param` = i: put info files recursively in subdirectories
- `param` = d: delete recursively .eps plots in all subdirectories
- `param` = g: restore recursively graphics in all subdirectories
- `param` = u: undo move of rejected data files to `_discarded` subdirectory
- `param` = 3: simulation of a correlation function + formatting
- `param` = 4: processing of a simulated function that is already formatted for CONTIN.

V. OUTPUT FILES

- All "TMP_*" files as well as the scripts "o", "f", "r" et "z" are updated at each run. They can be deleted without problem. Some files contain interesting informations that can be used for custom plots:

TMP_C: text file of the output by CONTIN (old-fashioned listing !)

TMP_CORREL_INIT: initial data $y(t)$

TMP_CORREL: preformed data $y_a(t)$ (cf. analysis row in the table of section VII)

TMP_DISTRIB: distribution function of relaxation times

TMP_VERIF: t , $y_a(t)$, calculated $y(t)$, point number, residuals

TMP_PARAM et TMP_FITFCT: fit input values, fit function and range of fit

The files beginning with the input filename are those that will probably be kept by the user:

filename_d.txt: distribution function output by CONTIN

filename_fit.txt: fit on the distribution function output by CONTIN

filename_p.eps: plot of the data and distribution function

The command `r` rename some temporary ".eps" files to facilitate their conservation.

- The "*.eps" plot files can be opened, printed or imported in reports.

- Batch mode: the useful output files are renamed and put into subdirectories according to their contents. The temporary files are deleted, a PDF aperçu file and a text file containing the statistics on the fits are created.

- Selection mode: updates statistics and aperçus with the data and plots corresponding to .ASC

files kept in the _ASC subdirectory, after the user has moved the unwanted .ASC files in the _discarded subdirectory. All the results corresponding to the latter are deleted except the original data.

VI. POSSIBLE CUSTOMARIZATIONS

- The input data files (NB: filename without extension) have a very precise format and are produced from original data, which can have very different origins or formats. In the present package, the Perl script `transform.pl` reads .ASC files output by the ALV-5000/E or ALV-7002 correlator softwares and formats them appropriately for CONTIN. This script must be modified (at the level of its subroutine `read_file`) if data have a different source (e.g. a different correlator or a rheometer). Similar changes will be necessary in the script `sum_ASC.pl`, which reads .ASC ALV files to calculate the average correlation function. For consistency the scripts `sum_ASC.pl` and `sim.pl`, which save outputs in an ALV-like format should also be modified in the output subroutine `output_ASC`.
- For filenames the following conventions are followed in the scripts:
 - Original data files have an extension .ASC but this extension is never typed.
 - The input files formatted for CONTIN are given the same name with extension dropped.
 - Temporary files begin always with "TMP_".
- CONTIN default input parameters: to display in the same plot the correlation function and the distribution function, CONTIN is parametrized to work with a distribution function of relaxation times. This can be modified (see CONTIN documentation in /Applications/CNTb/docs, "contin-manual1" et "contin Update2") but most graphical outputs will have to be worked out again !

VII. INPUT PARAMETERS

- The input parameters are detailed in the original papers by S.W. Provencher and the user's manual "contin-manual1". They control the way CONTIN processes the data. As a quick reminder:

- RUSER10 sets the preforming of data $y(t)$ to feed CONTIN
 - = 0: input = y (no modification)
 - > 0: input $y_a = \sqrt{y/RUSER10-1}$
 - < 0: input $y_a = \sqrt{y}$

In the last two cases, negative arguments z yield $-\sqrt{-z}$ input values to prevent a bias in the baseline.

- NLINF sets the number of free baselines
 - = 0: fixed baseline (set to 0)
 - = 1: 1 fitted baseline
- IWT sets the weight $1/\sigma^2$ of data points in the calculus of the residuals

- = 1: σ^2 constant
- = 2: $\sigma^2 = \gamma$
- = 3: $\sigma^2 = \gamma^2$
- = 4: reads $1/\sigma^2$
- = 5: σ^2 calculated by USERWT. Default corresponds to DLS data in CONTIN.

- IUSER10 sets the type of distribution in use $F(\lambda, t) = f \lambda^{R23} \exp(-R21 + \lambda^{R22})$
 - = 1: $R23 = 1$ $R21 = R18 q^2$ if $D=R18 M^{R22} \Rightarrow P(M)$
 - = 2: $R23 = 0$ $R22 = 1$ $R21 = q^2 \Rightarrow$ Laplace transform ($R16=0$), $P(D)$ ($R16 \neq 0$)
 - = 3: $R23 = 3$ $R22 = -1$ $R21 = k_B T q^2 / (6\pi\eta) \Rightarrow P(R)$
 - = 4: $R21, R22, R23$ input by user ! if $P(\tau)$ wanted then $R21 = 1, R22 = -1, R23 = 0$

The input parameters should a priori be the following for a time analysis of $\gamma(t)$ original data, depending of their origin and the desired preforming.

Definitions for RUSER18, RUSER19 and RUSER22 depend on IUSER10.
RUSER20 is calculated from the other input values.

The scripts in the package define correctly these parameters without any action by the user.

$\gamma(t)$	$G^{(2)}$	$g^{(2)}$	$g^{(2)}-1$	$g^{(1)}$	$G_R(t)$
$\gamma_a(t)$ analyzed	$\gamma : (\gamma/R10-1)^{1/2}$	$\gamma : (\gamma-1)^{1/2}$	$\gamma : \gamma^{1/2}$	γ	γ
RUSER10	0 ; >0	0 ; 1	0 ; <0	0	0
NLINF	1 ; 0,1	1 ; 0,1	0,1 ; 0,1	0	0
IWT	5 ; 5	5 ; 5	5 ; 5	2	1
IUSER10	4	4	4	4	4
RUSER15	refractive index n_D	n_D	n_D	n_D	-
RUSER16	wavelength (nm) λ	λ	λ	λ	-
RUSER17	angle θ	θ	θ	θ	-
RUSER18	temperature (K) T	T	T	T	-
RUSER19	viscosity (cP) η	η	η	η	-
RUSER20	scattering wavevector q (calc.)	q	q	q	-
RUSER21	1	1	1	1	1
RUSER22	-1	-1	-1	-1	-1
RUSER23	0	0	0	0	0

VIII. FORMAT FOR FILES [INFO_C](#), [INFO_F](#) and [INFO_P](#)

```
#####
# format for INFO_C containing info for CONTIN
#####
1e-6          # first point [s]
10            # last point [s]
1e-5          # Tau min [s]
.1            # Tau max [s]
100           # Number of grid points
-1            # integer: 1 if $1=2 , -1 if $1=1
1             # integer: 1 if free baseline, 0 if fixed baseline
5             # integer: 5 if weighted residuals, 1 if unweighted

#####
# format for INFO_F containing info to fit distribution of relaxation times
#####
2             # number of peaks
-5            # log(tmin [s])
-1            # log(tmax [s])
a1=.1         # a1=amplitude of first peak
b1=-4         # b1=log(position [s]) of first peak
c1=.3         # c1=log("width") of first peak
a2=.1         # a2=amplitude of second peak
b2=-3         # b2=log(position [s]) of second peak
c2=.3         # c2=log("width") of second peak
# ...         # etc... if more peaks
```

It is possible to bypass the need to feed the fitting script with initial free parameter values through [INFO_F](#). Instead an automatic search for the peaks in the distribution is done with an algorithm using second derivative method.

To use this feature simply leave a blank input line in the place of the entry "a1=..." in [INFO_F](#). Then there are two options according to the number of peaks N_p entered in the first line:

- If N_p is an integer larger than 1 ($N_p \geq 1$), the script will select the N_p larger maxima in the time distribution and start with these initial values.
- If N_p is smaller than 1 ($0 \leq N_p < 1$), the script will fit a variable number of peaks. The value N_p serves then as a criterion to discard small peaks discovered by the algorithm, which have a contribution smaller than about N_p times the one of the major peak in the distribution. When N_p is set to 0, then the criterion is set by default to 1%, i.e., $N_p = 0$ is equivalent to $N_p = 0.01$.

With $N_p < 1$, the user might end up with a varying number of peaks among the different measurements. Then the averages performed in the statistical analysis of the results are meaningless and are skipped. Only the results for the average correlation function are listed. A warning is issued in the results file.

```
#####
# format for INFO_P containing info to set criteria for data rejection
#####
0.2                # maximum value for  $\Delta I / \langle I \rangle$ 
0.01               # maximum value for asymptotic  $g^{(1)}(t)$ 
1                  # maximum value for  $g^{(2)} - 1$ 
```

The above values are those that will be used if no [INFO_P](#) file is present. The files for which these values are exceeded are automatically moved to the subdirectory [_discarded](#) but the corresponding raw data plots are nevertheless shown in the pdf file created when the option "p" is used for [cnt](#). The automatic discarding of data files is done only with this option. The discarded files can be moved back by typing: [cnt dirname u](#)

NB: in the above INFO_ files do not enter the right-most part of the lines starting with #. These are comments to give a hint for the use of the corresponding value.

IX. INSTALLATION DIRECTIVES (OS X)

OS X.5 and OS X.6

- Proceed in the following order that was tested successfully.
- If [X11](#) is not present in directory Applications/Utilitaires:
 - Insert and open the installation DVD of system X.5, X.6 or X.7.
 - Open the directory [Installations facultatives \(Optional installs\)](#).
 - Double-click [Optional Installs.mpkg](#).
 - Answer the usual questions.
 - When in the step '(Installation type)', click on the triangle Applications.
 - Tick X11
 - Continue the installation.
- To install [Gnuplot](#):
 - Visit [sourceforge site](#)
 - Download Gnuplot-4.4.0.tar.gz and decompress it.
 - Launch Terminal and open a terminal window (menu Applications)
 - Change working directory with [cd](#) command until [inside](#) Gnuplot-4.4.0 directory.
 - Enter the successive commands:


```
./configure --with-readline=bsd --x-include=/usr/include/X11 --x-libraries=/usr/X11/lib
make
sudo make install
```

 and enter the requested administrator password.
 - Quit and launch X11.
 - Enter [Gnuplot](#). If everything is fine, a prompt [Gnuplot>](#) shows that Gnuplot is running.

When entering the command **test** a graphic window pops up and shows a sample of available graphic capabilities. Close it and enter **quit** to quit Gnuplot.

- To install **fortran** libraries compatible with the fortran 66 used in CONTIN, e.g.:
 - Visit this page of [sourceforge](#) site to download the GCC GNU compiler corresponding to your system and processor.
 - Open a terminal window in X11 and follow their instructions.

OS X.7

No experience with this one but the installation should likely proceed similarly to previous system versions.

OS X.8

X11 is no longer provided in OS X.8 and Apple computers tend now to the same autism as their smartphones. You have to download instead XQuartz, e.g., [here](#).

Many complaints have arisen against the increasing difficulties to install Gnuplot in this system (see e.g. [here](#)). A possible simple workaround is given in this [blog post](#) but I did not try it. Instead I followed the procedure detailed [here](#) using Gnuplot-4.6.1 downloaded from [sourceforge site](#). It worked in my case (Macbook Pro Retina, OS X.8.2). To install fortran libraries proceed like for the previous system versions.

X. OTHER OPERATING SYSTEMS

CNTb should also run on any Windows or LINUX operating systems providing the following requirements are met:

- Perl and Gnuplot should be installed on your computer with the appropriate procedure.
- If a FORTRAN executable of CONTIN has to be generated for your processor, you will need a FORTRAN compiler. S.W. Provencher provides two executables for Windows and LINUX users that might run on a number of usual configurations. The source code for CONTIN can be downloaded at the same [address](#). It is also included in directory **CNTb/for** for convenience. The executable should be placed in directory **CNTb/bin** and named **contin**.
- For versions 3 and earlier, the path names should be adapted to your operating system and configuration. In particular, they should be changed in the following scripts:

```
CNTb/cnt
CNTb/bat/cnt1by1
CNTb/bat/cntb
CNTb/bat/intb
CNTb/bat/select
CNTb/bat/refit
CNTb/bat/init
```

These text files can be edited with a text editor and a search and replace command. You should simply replace the string `"/Applications"` by the appropriate path for your directory structure. Command lines should then be entered according to your modifications.

Starting June 2013, version 4 does no longer require this update of path names.

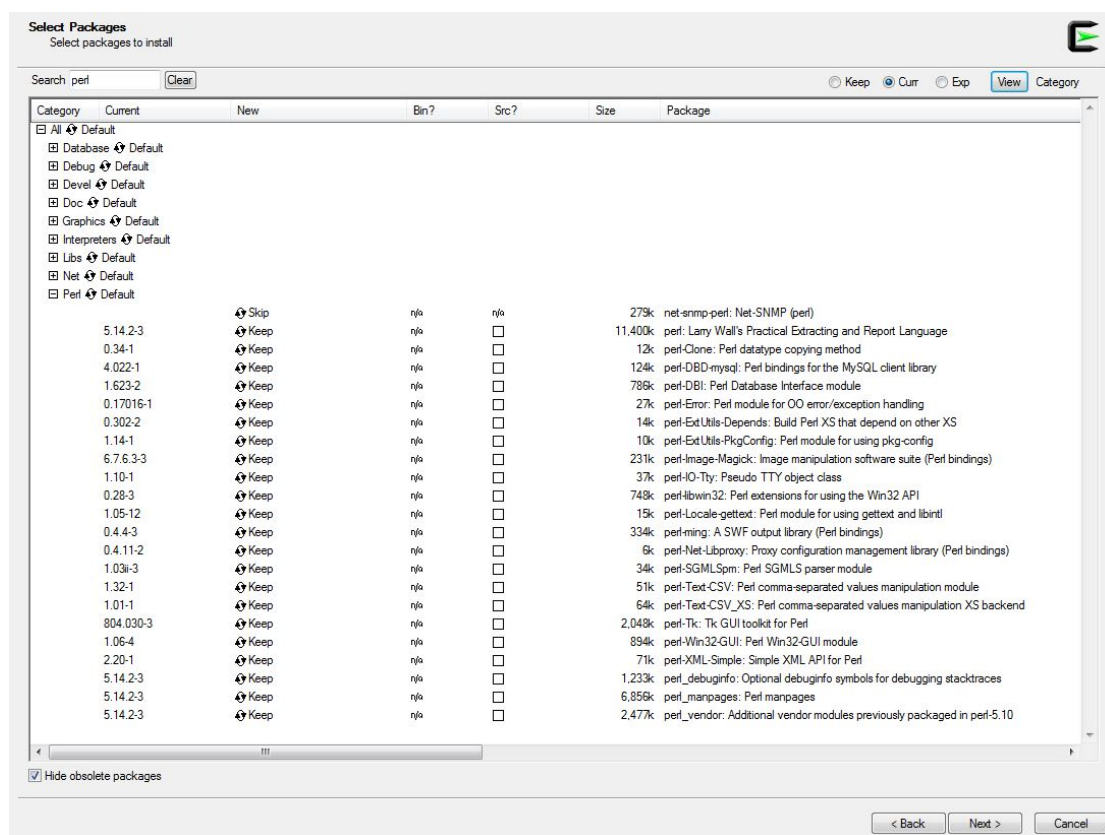
• Windows users have two options: they can either choose to use VirtualBox and install a Linux system, or download the free X server designed by [Cygwin](#) and type the CNTb command lines in a X terminal window opened with Cygwin/X. Both solutions create a new hierarchy tree of files that cannot communicate easily with the Windows system.

The VirtualBox solution offers the advantage of easy installation of Gnuplot, Perl, and FORTRAN compiler in the appropriate places. The LINUX package I used was Ubuntu. By default, the utility [pstopdf](#) is not present in this package and has to be installed afterwards. Some graphic outputs are still misaligned with this solution and need some improvement (todo list).

The Cygwin solution is also quite easy to install with the following steps. However it is much slower for the production of PDF files. So if you plan to analyze large numbers of files, you should prefer the first solution. [Here](#) version x86 running on Windows7 64 bit was tested. Double click the downloaded Installer, select root and installation directories, click on next until the Select-Packages step.

The required packages are Perl, Gnuplot and Xinit. Find each of them by using the search bar and install them by clicking on the line under the column **New** until **Keep** or the version number appear.

Install the package "[Perl: Larry Wall's practical Extracting and Report Language](#)"

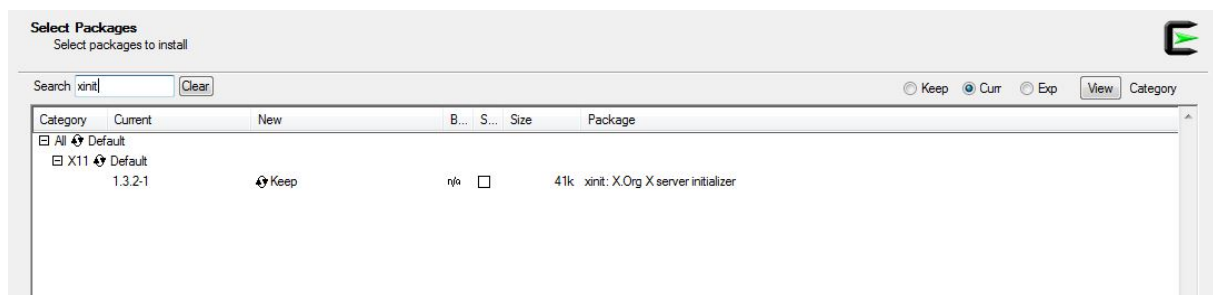


version 4 2013/12/12

Install the package "Gnuplot: A command line driven interactive function plotting utility"

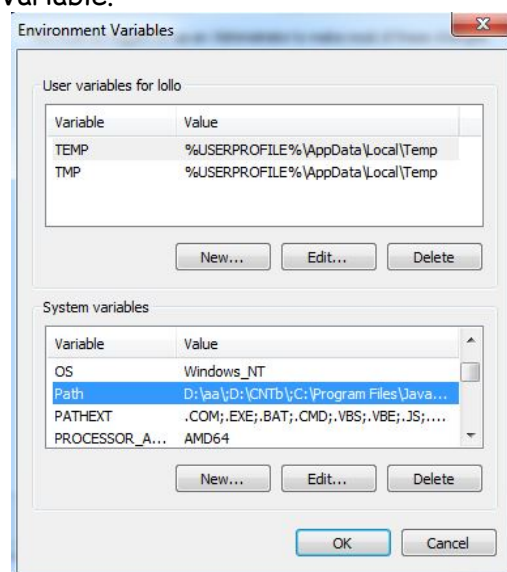


Install the package "xinit: X Org X server initializer"



Once all these packages have been selected, click on next and wait for the installation to finish. To check if all the packages were installed you can run again the installer and check that all the packages shown in the pictures above are displayed as **Keep** just like in the screenshots.

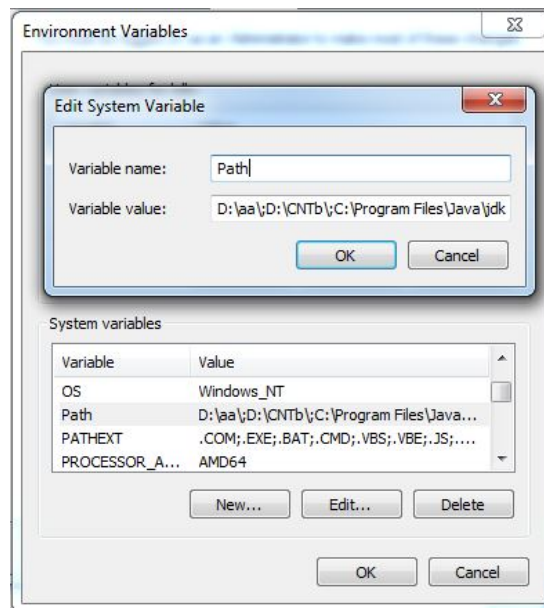
Extract the folder **CNTb** in any directory and add its path to the PATH variable:
For Windows 7, in **Control Panel** open **System**, click on **Advanced System Settings** and click on **Environment Variables** in the bottom of the window. Once there, scroll down under **System Variables** until you see the **Path** Variable.



Click on it twice and add the path to the location of CNTb. In this case the path was D:\CNTb\.

All path variables must be separated by a ";" symbol.

WARNING: DO NOT DELETE ANYTHING IN THE VARIABLE VALUE !!!



Install [Ghostscript](#) (tested using the 32 bit version).

The bin file to run Contin needs to be the Windows version. It can be compiled using [gfortran](#) from the window commandline:

Open the start menu and run [cmd.exe](#)

Go to the directory where CNTb was extracted: move to the correct drive by typing the drive letter followed by a semicolon, e.g., [E: <return>](#), then reach the CNTb folder by typing [cd](#) followed by a space and the complete path to the CNT\for\CONTIN directory, e.g., [E:\>cd CNTb\for\CONTIN <return>](#).

Type [gfortran contin.for <return>](#) and wait for the compilation to be completed. At the end you should have created a file called [a.exe](#), which needs to be moved to the bin folder and renamed [contin](#) without any extension. You can do this by typing: [copy a.exe contin <return>](#). This will create a file called [contin](#) without an extension, which can then be moved to the bin folder.

Run [Xwin Server](#). From the terminal window you can now use CNTb as described above.

XI. UPDATES

2013 June

- Release of version 4. All the main shell scripts have been rewritten in Perl, which allows the use of the FindBin module in the Standard Perl Library in every place where necessary. Thus there is no longer a need to update paths according to a specific user's configuration and the CNTb folder can be placed anywhere.

2013 March

- Release of version 3. Peak fits by Gnuplot are automatically discarded if quality criteria are not met. For technical reasons, scripts used to fit and plot the results had to be gathered in a single large script. The single script "fit" replaces previous "fit", "fitparam", "fitshow" and "getfit" scripts. Errors on the fit are now traced and can be used for further propagation, e.g., in the calculation of the peaks' area.

2013 February

- Release of version 2. CNTb becomes fully recursive and there is no longer a need to distinguish between batch, super or hyper batch modes. The level of recursions is not limited.
- Possibility to automatically discard bad data. Criteria to define data quality set in optional file [INFO_P](#). Default values used when running "[cnt dirname p](#)" are given in VIII. Discarded data can be restored to their initial place by running simply "[cnt dirname u](#)".
- All scripts in CNTb/bat have been made executable and renamed without .pl extension.
- Fortran format for the relaxation times in [INFO_C](#) is no longer necessary.

2013 January

- New possibility to bypass the need to give initial values for the free parameters in the fitting procedure of the time distribution peaks and to generate an automatic search of these peaks (see VIII).
- Fixed some parts of the scripts that caused messy outputs when Gnuplot did not converge properly in the fit of the time distribution. Now in these cases the results are produced silently with the initial free parameter values. However a warning is issued in the results file and the plot of the fit gets a label "BREAK in the fit".

2012 December

- New possibility to fit the distribution function without running again CONTIN in batch mode as formerly available in interactive mode only.
- New possibility to produce a pdf file with the plots of intensity traces and correlation functions. Thus raw data can be easily screened without performing CONTIN analysis.

2012 Jan.

- Support added for ALV 7002 data files format.

Bugs can be reported to francois.schosseler@ics-cnrs.unistra.fr. Comments are also welcome at the same address. Updates can be downloaded on the web site of Institut Charles Sadron [here](#). The Institut Charles Sadron is a laboratory of the Centre National de la Recherche Scientifique.